

Emmanuel Gnabeyeu Mbiada

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Profile

Doctoral Researcher in Applied Mathematics, and Artificial Intelligence at Sorbonne Université (LPSM), passionate about bridging theory and applications. Research interests include Reinforcement Learning, Stochastic Analysis, Stochastic Optimal Control, Stochastic Optimization for Machine Learning, Bayesian Statistics, Langevin Score's based Diffusion Models, Probabilistic Modelling, and Probabilistic Machine Learning.

Academic Background

- 2024–2027 **Ph.D. Researcher in Mathematics and Applications** — Sorbonne Université & Université Paris Cité, Laboratoire de Probabilités, Statistique et Modélisation, Paris, France.
- **Thesis:** *Volterra Equations, Asymptotic Stationarity and Applications.*
- 2023–2024 **MPhil/M.Sc. in Applied Mathematics and Statistics (Track Probability and Finance)** — Sorbonne Université & Institut Polytechnique de Paris, France. *Highest Honours.*
- *Machine Learning and statistical models for multiscale market finance.*
 - **Thesis:** *Reinforcement Learning for Volatility Fitting.*
 - **Honors:** *Summa Cum Laude, Outstanding Dissertation Award*
- 2020–2024 **Ingénieur Polytechnicien (Master level)** — École Polytechnique (I'X), Palaiseau, France.
- *Applied Mathematics, Computer Science and Economics.* *With Distinction.*
- 2021–2022 **Bachelor's Degree in Philosophy** — Université Paris Nanterre, France.
- *Politics, Philosophy and Theology.*
- 2017–2020 **B.Sc. in Mathematics** — Université de Yaoundé I, Cameroon.
- *High Honours; Applied and Computational Mathematics.*
- 2017–2020 **Preparatory Classes (Mathematics & Physics)** — Ecole Nationale Supérieure Polytechnique de Yaoundé (ENSPY), Yaoundé, Cameroon.
- *Mathematics and Physical Science.* *Highest Honours.*

Selected Publications and Preprints

- July 10, 2026 **Gnabeyeu, E. & Souilmi, S.** Smile Dynamics under the fake stationary quadratic rough Heston Volatility: SPX and Skew-Stickiness Ratio, in preparation.
- June 30, 2026 **Gnabeyeu, E. & Pagès, G.** Fake Stationarity in Stochastic Volterra Integral Equations with Affine Drift and Long-Memory Kernels, in preparation.
- May 10, 2026 **Gnabeyeu, E. & Dro, S. B.** Optimal Merton's Problem under Multivariate Affine Volterra Models with Jumps.
- Avril 1, 2026 **Gnabeyeu, E.** On the mean-variance problem through the lens of multivariate fake stationary affine Volterra dynamics.
- March 22, 2026 **Gnabeyeu, E. & Pagès, G.** On Path-dependent Volterra Integral Equations: Strong Well-posedness and Stochastic Numerics, *Submitted for publication.*
- March 11, 2026 **Gnabeyeu, E.** On Utility Maximization under Multivariate Fake Stationary Affine Volterra Models, *Submitted for publication.*

- February 11, 2026 **Gnabeyeu, E. & Pagès, G. & Rosenbaum, M.** Fake stationary rough Heston volatility: Microstructure-inspired foundations, *Submitted for publication*.
- November 1, 2025 **Gnabeyeu, E. & Pagès, G. & Rosenbaum, M.** On Inhomogeneous Affine Volterra Processes: Stationarity and Applications to the Volterra Heston Model, *Submitted for publication*.
- September 1, 2025 **Gnabeyeu, E. & Pagès, G.** On a Stationarity Theory for Stochastic Volterra Integral Equations, to appear in *Stochastic Processes and their Applications*.
- October 22, 2024 **Gnabeyeu, E., Karkar, O., Idboufous, I.** Solving the Dynamic Volatility Fitting Problem: A Deep Reinforcement Learning Approach, to appear *Journal of Machine Learning Research*.

Conferences and Summer Schools

- June 29th – July 3rd, 2026 **XIII Bachelier World Congress of the Bachelier Finance Society 2026** — University of Bologna, Bologna, Italy.
- *Attended plenary talks by a distinguished lineup of international academics, practitioners, and researchers, covering recent advances and key emerging topics in financial mathematics.*
- 27–30 January 2026 **Conference on Advances in Financial Mathematics 2026 : directions and some information** — Sorbonne University, Campus Pierre et Marie Curie, Paris, France.
- *Attended plenary talks on recent advances in financial Mathematics, from both academic researchers and practitioners.*
- June–July 2025 **International Joint Conference on Neural Networks (IJCNN 2025)** — Pontifical Gregorian University, Rome, Italy.
- *Presenting the paper "Solving the Dynamic Volatility Fitting Problem: A Deep Reinforcement Learning Approach in Continuous State Action Spaces."*
 - *Attended workshops on Reinforcement Learning, Diffusion Models and Flow, Deep Generative Models, and Bayesian Methods for Inference and Learning.*
- July 2025 **5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance** — Centre de Recerca Matemàtica (CRM), Autonomous University of Barcelona, Spain.
- *Participated in lectures on rough volatility and rough path theory, and their applications to signatures, machine learning, and stochastic optimal control.*

Selected Talks

- June 30th, 2026 **13th World Congress of the Bachelier Finance Society** — University of Bologna - Bologna, Italy.
- *On the fake stationary rough volatility models.*
- June 2026 **Bachelier Seminar** — Institut Henri Poincaré - Paris, France.
- *A Journey Through Exponential Utility Maximization under Multivariate Fake Stationary Affine Volterra Models.*
- December 2025 **Doctoral Seminar** — Sorbonne Université, LPSM, Paris, France.
- *On Inhomogeneous Fractional Square-Root Process: Microstructural Foundation and Weak Stationarity Theory.*
- October 2025 **Natixis Foundation for Research and Innovation Awards Ceremony for Best Master's Thesis in Quantitative Finance** — Sky Garden (BPCE Tower), Paris, France.
- *A Reinforcement Learning Perspective to Exotics and Flow Equity Derivatives.*
- July 2025 **5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance** — Centre de Recerca Matemàtica, Spain.

- *On a Stationarity Theory for Stochastic Volterra Integral Equation with Affine Drift: Finite-Time, Functional Weak Asymptotics and Applications.*

Teaching Experience

- 2025–May. 2026 **Teaching Assistant** — Sorbonne Université, Paris, France.
- *Course: Numerical Analysis (Bachelor's Level 3, Mathematics).*
 - *Course: Advanced Mathematics (Bachelor's Degree – First Year, Mathematics).*
- Sept.–Dec. 2025 **Teaching Assistant** — Université Paris 1 Panthéon-Sorbonne, Paris, France.
- *Course: Probabilities and Statistics (Bachelor Level 3, Economics).*
- Sept.–Dec. 2025 **Teaching Assistant** — ESILV – Graduate Engineering School, Paris, France.
- *Course: Advanced Statistics (Master 1).*
 - *Covered topics: Inferential and Bayesian Statistics, Non-parametric Statistic and Hypothesis Testing.*
- 2018–2020 **Teacher and Tutor** — Intelligencia Corporation, Yaoundé, Cameroon.
- *Tutoring in mathematics and physics.*

Professional Experience

- Apr.–Nov. 2024 **Quantitative Researcher** — Citigroup Global Markets Limited, London, UK.
- *Machine Learning for flows and exotic equity derivatives.*
- Apr.–Sept. 2023 **Quantitative Researcher & Strategist** — BNP Paribas Asset Management, Paris, France.
- *Machine Learning for mid-frequency systematic equity investment strategies.*
- Jun.–Sept. 2022 **Data Scientist & ML Engineer** — BOXY Start-Up, Paris, France.
- *Mathematical modelling and demand forecasting in autonomous retail systems.*
- 2021–2022 **Data Scientist Intern** — Alvidence Start-Up & École Polytechnique, Palaiseau, France.
- *Explainability of AI and statistical learning algorithms (XAI project).*

Awards, Honors, Distinctions and Scholarships

- 2025 **First Prize – Best Master's Thesis in Quantitative Finance** — Natixis Foundation for Research and Innovation.
- *First Prize for the 2025 edition of the best master's thesis in Mathematical Finance.*
- 2021 – 2023 **Eiffel Excellence Scholarship.** — from the French Government.
- 2017 **Diploma of Excellence – National Scientific Olympiad** — Cameroon.
- 2015 – 2017 **Special Honour Roll for Best Laureate in Mathematics and Physics.** — NIAT Foundation.
- 2011 **Africa Challenges Group Merit Scholarship.** — from the United States.

Skills

- **Programming:** Python, Java, SQL, MATLAB, JavaScript, R, C/C++, PHP, CSS.
- **Libraries:** Pandas, Scikit-learn, PyTorch.
- **Languages:** French (Native), English (C1), German (B1), Spanish (A2).

Interests

- Reading (literature, philosophy, politics, theology).
- Travel, music, cinema, theatre, drawing, boxing.